

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

STRUCTURAL ORGANISATION IN ANIMALS (Frog)

LECTURE-1

ZOOLOGY

DAVOOD SIR (CEO of NG)



Topics To Be Covered

- 1** Morphology of a frog.
- 2** Anatomy of a frog
 - a** Digestive System of a Frog
 - b** Previous Year NEET MCQ'S

Introduction the Chapter

- ① Morphology:- Study of external visible characters/features
- ② Anatomy:- Study of Structures of internal organs.

Frogs



Phylum:- Chordata

Class:- Amphibia

Rana tigrina:- (Indian Frog).

Cold blooded or poikilotherms:- their body temperature varies with the temperature of the Environment.

Camouflage:- they have the ability to change the colour in grasses on dry land to hide them from their enemies. this phenomenon is called Camouflage.

Mimicry:-> H/W.

Summer sleep (aestivation) and winter sleep (hibernation):- During peak summer and winter frogs take shelter in deep burrows to protect them from extreme heat & cold.

Frogs

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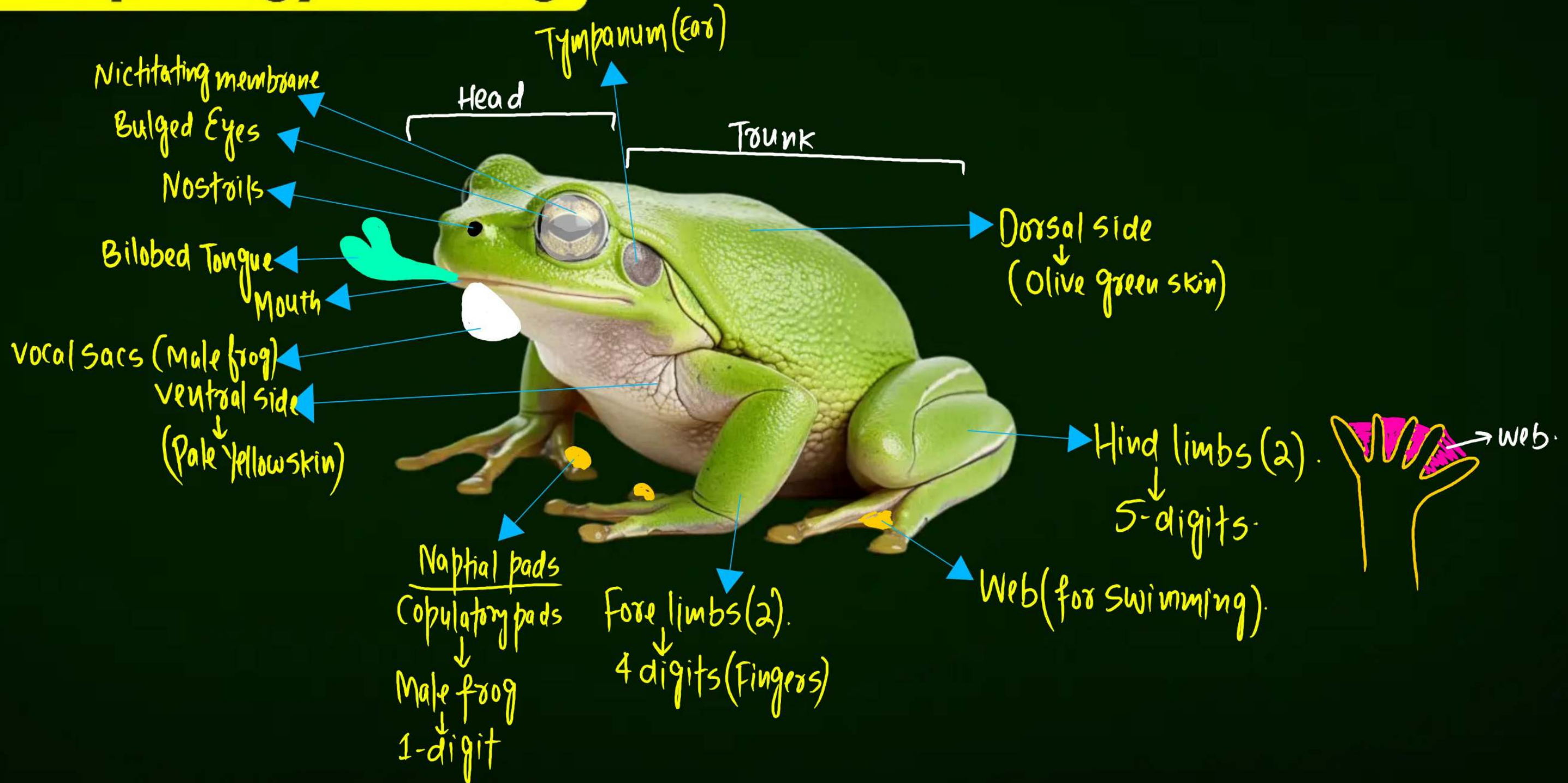
Rana tigrina (Indian Frog).

Cold blooded or poikilotherms:- Their body temperature varies with the temperature of the environment.

Camouflage:- They have the ability to change the colour in grasses and on dry land to hide them from their enemies. This protective coloration is called **mimicry**.

Summer sleep (aestivation) and winter sleep (hibernation):- During peak summer and winter, they take shelter in deep burrows to protect them from extreme heat and cold.

Morphology of a Frog



Morphology of a Frog

- ✓ **Skin:-** Smooth, Slippery & moist with mucus
- ✓ **Body Surface:-** ventral side have pale yellow skin & Dorsal side have Olive green skin.
- ✓ Not drinking water but absorbs through skin. (osmosis).
- ✓ **Body:** is divided into Head & Trunk. (Neck & Tail is Absent).
- ✓ Nostrils are present above the mouth.
- ✓ **Eyes:-** Bulged eyes with protective nictitating membrane protects them in water.
- ✓ **Tympanum (ear):-** On either side of eyes and receive sound
- ✓ **Limbs:** Forelimbs (4 digits), hind limbs (5-digits). Hind limbs are muscular & feet have webbed digits that help in swimming. Fore limbs have Nuptial pads on first digit for copulation
- ✓ **Sexual dimorphism:-** (Sexes are separate means males and females are present).
- ✓ **Male frogs:-** Vocal Sacs (for Croaking sound to attract females) and Nuptial pads on first digit of fore limb.

Anatomy of a Frogs

✓ Body cavity of frogs: Contains

① Digestive System. ✓

② Respiratory System.

③ Circulatory System

④ Nervous System.

⑤ Excretory System

⑥ Reproductive System.

Question: NEET-2025

4-Marks

Which of the following statement is correct about location of the male frog copulatory pad?

- A** First digit of hind limb
- B** Second digit of fore limb
- C** First digit of the fore limb.
- D** First and second digit of fore limb



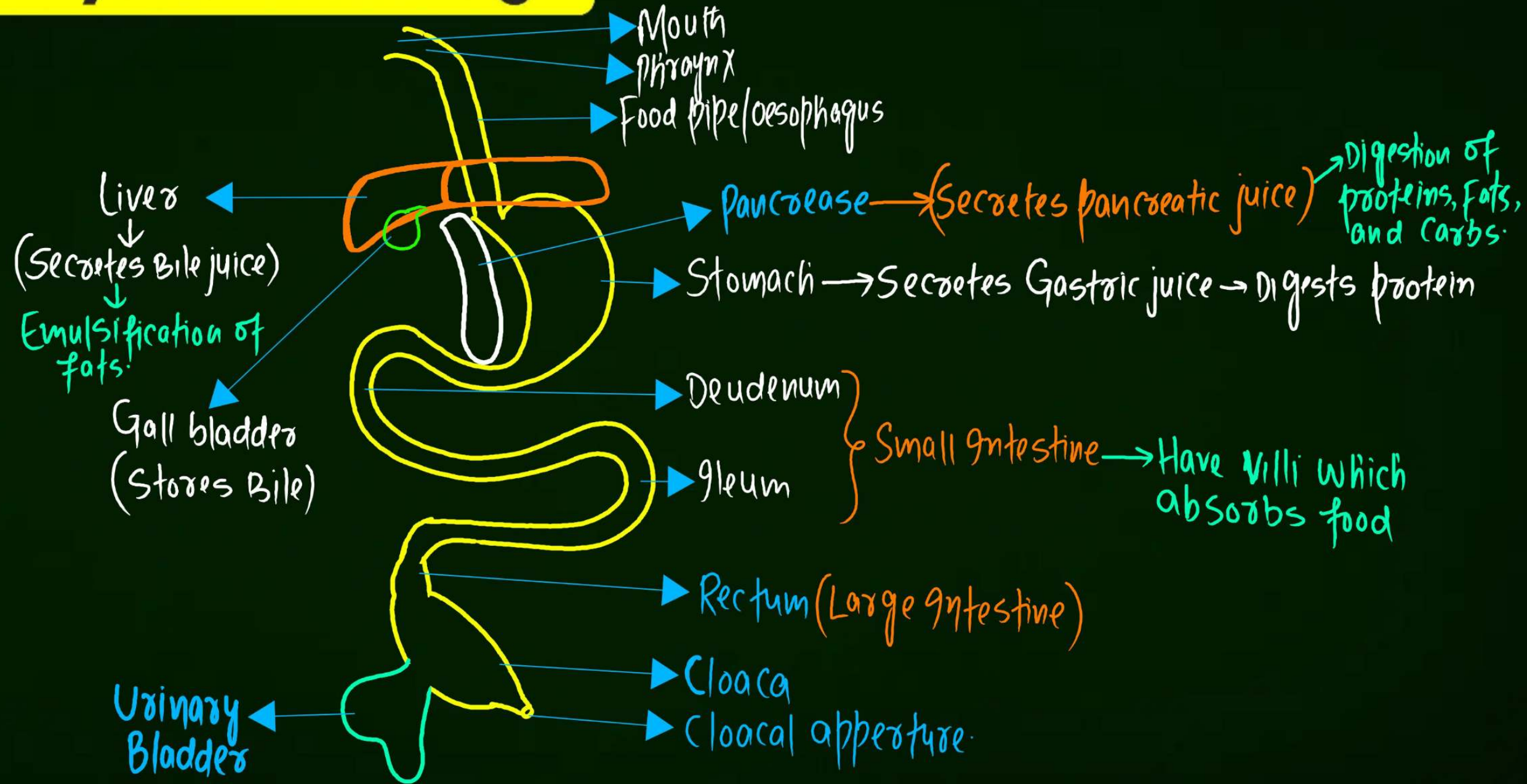
Question: NEET-PYQ4-Marks

Which of the these is not a characteristic feature of a frog?

- A** Webbed Feet
- B** Bilobed Tongue.
- C** Thick Leathery skin.
- D** Membranous Tympanum



Digestive system of a Frog



Digestive system of a Frog

- ✓ The digestive system includes the alimentary canal and digestive glands.
- ✓ The alimentary canal is short because frogs are carnivores with a reduced intestine length.
- ✓ Frogs capture food with a bilobed tongue.
- ✓ The mouth leads to the buccal cavity, then to the oesophagus via the pharynx.
- ✓ The short oesophagus opens into the stomach, which continues as the intestine, rectum, and finally the cloaca.
- ✓ The liver secretes bile stored in the gall bladder.
- ✓ The pancreas produces pancreatic juice containing digestive enzymes.
- ✓ Digestion begins in the stomach with HCl and gastric juices.
- ✓ Partially digested chyme moves to the duodenum.
- ✓ Duodenum receives bile (from gall bladder) and pancreatic juice (from pancreas) via the common bile duct.

Digestive system of a Frog

- ✓ Bile emulsifies fats; pancreatic juice digests carbohydrates and proteins.
- ✓ Final digestion occurs in the intestine.
- ✓ Villi and microvilli in the intestine absorb digested food.
- ✓ Undigested waste moves to rectum and exits via cloaca.

Thank
You

